Johan Ajnabi

PhD Candidate | Epithelial Biology, Mechanobiology and Epigenetics

BRIC-Institute for Stem Cell Science and Regenerative Medicine (inStem)
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Summary

Mechanobiology and epigenetics researcher investigating the mechanistic basis by which mechanical forces regulate chromatin organization and cell fate transitions during tissue repair. Identified a mechanosensitive regulatory mechanism controlling DNMT3a nucleocytoplasmic partitioning and linking cytoskeletal dynamics to epigenetic reprogramming. Integrates quantitative live-cell imaging, molecular perturbation, and cell biology to define how tissue architecture governs regeneration and disease.

Education

PhD Candidate, BRIC-Institute for Stem Cell Science and Regenerative Medicine (inStem) — *2019 - Present*
Advisor: Prof. Colin Jamora
PhD thesis: Regulation of DNMT3a localization in the cutaneous wound healing response

M.Sc. in **Molecular Biology and Biotechnology**, ICAR-Indian Agricultural Research Institute — *2019*

B.Sc. (Hons.) in **Agriculture**, Bidhan Chandra Krishi Viswavidyalaya — *2017*

Research Experience

BRIC-Institute for Stem Cell Science and Regenerative Medicine (inStem), India

2019-Present

Research Scholar (PhD)
Advisor: Prof. Colin Jamora

Mechanotransduction and epigenetic regulation (Primary project)

- Identified a mechanosensitive regulatory mechanism controlling DNMT3a nucleocytoplasmic partitioning and linking cytoskeletal dynamics to epigenetic reprogramming during epidermal wound repair.
- Integrated live-cell imaging, biochemical assays, and genetic perturbations to define ERK-dependent mechanotransduction pathways
- Led first-author study (bioRxiv: Ajnabi *et al.*, 2026).

Stem cell signaling and niche regulation

- Contributed to identifying a Mindin-integrin-STAT3 signaling axis that sustains keratinocyte stemness through integrin endocytosis (Cell Communication and Signaling: Dam *et al.*, 2026).
- Contributed to work showing how autocrine Mindin signaling maintains stem and progenitor states in skin epithelium and carcinoma cells (Cell Reports: Badarinath *et al.*, 2022).

Tissue-scale and translational biology (vitiligo, fibrosis)

- Performed spatial transcriptomic profiling of vitiligo skin before and after NB-UVB treatment and validated immune-state changes by immunofluorescence and qPCR (Clinical & Translational Immunology: Dutta *et al.*, 2026).
- Studied extracellular matrix regulation of cutaneous fibrosis using transgenic mouse models (Journal of Investigative Dermatology: Rana *et al.*, 2023).

Host-pathogen interactions

- Investigated LL-37-mediated inhibition of SARS-CoV-2 entry using mammalian cell systems, pseudovirus assays, and flow cytometry (Frontiers in Immunology: Bhatt *et al.*, 2023).

ICAR-National Institute for Plant Biotechnology, India

2017-2019

M.Sc. dissertation
Advisor: Dr. Monika Dalal

- Characterized promoter cis-regulatory regions controlling *PM19* expression in wheat using molecular cloning, transient GUS assays, plant tissue culture, transformation, and sequence analysis (Thesis: Ajnabi J, 2019)

Publications

First-Author & Lead Contributions

Ajnabi J, Dam B, Gupta E, Saha T, Dutta A, Kumar S, Gupta A, Palakodeti D, Jamora C. *Actin-dependent mechanotransduction controls nucleocytoplasmic partitioning of DNMT3a through ERK1/2 signaling during cutaneous wound healing.* **bioRxiv**, 2026 *(preprint)*

Collaborative Publications

Dam B, **Ajnabi J**, Saha T, Shrivastava A, Badarinath K, Hegde A, Dutta A, Jasoria S, Kataria S, Singh A, Jamora C. *Mindin-mediated αM-integrin endocytosis activates STAT3 to maintain keratinocyte stemness.* **Cell Communication and Signaling**, 2026

Dutta A, Gupta D, **Ajnabi J**, Dam B, P S, Jamora C. *Spatial transcriptomic analysis of the immune landscape following NB-UVB treatment of vitiligo skin.* **Clinical & Translational Immunology**, 2026

Dutta S, Islam Z, Das S, Barman A, Chowdhury M, Mondal BP, **Ajnabi J**, Manna D. *Harmonizing plant resilience: membrane lipid dynamics in response to abiotic stresses.* **Discover Plants**, 2025

Bhatt T, Dam B, Khedkar SU, Lall S, Pandey S, Kataria S, **Ajnabi J**, Gulzar SEJ, Dias PM, Waskar M, Raut J, Sundaramurthy V, Vemula PK, Ghatlia N, Majumdar A, Jamora C. *Niacinamide enhances cathelicidin-mediated SARS-CoV-2 membrane disruption.* **Frontiers in Immunology**, 2023

Rana I, Kataria S, Tan TL, Hajam EY, Kashyap DK, Saha D, **Ajnabi J**, Paul S, Jayappa S, Ananthan ASHP, Kumar P, Zaarour RF, Haarshaadri J, Kansagara G, Rizvi A, Zirmire RK, Badarinath K, Khedkar SU, Chandra Y, Samuel R, George R, Danda D, Jacob PM, Dey R, Dhandapany PS, He YW, Varga J, Varghese S, Jamora C. *Mindin (SPON2) is essential for cutaneous fibrogenesis in a mouse model of systemic sclerosis.* **Journal of Investigative Dermatology**, 2023

Badarinath K, Dam B, Kataria S, Zirmire RK, Dey R, Kansagara G, **Ajnabi J**, Hegde A, Singh R, Masudi T, Sambath J, Sachithanandan SP, Kumar P, Gulyani A, He YW, Krishna S, Jamora C. *Snail maintains the stem/progenitor state of skin epithelial cells and carcinomas through autocrine Mindin signaling.* **Cell Reports**, 2022

Fellowships and Awards

- **Paeonia Foundation Travel Award** – Selected to present research at *Mechanobiology in Health and Disease Conference*, National University of Singapore, 2023
- **CSIR-NET Junior Research Fellowship (JRF)** (All India Rank 20) – National-level fellowship from the Council of Scientific and Industrial Research (CSIR), India – one of India's most competitive fellowship exams for PhD funding in science, 2019
- **ICMR Junior Research Fellowship (JRF)** – Competitive national fellowship in Life Sciences from the Indian Council of Medical Research (ICMR), 2019
- Also qualified for five additional national-level fellowships/examinations: ASRB-NET (Agricultural Biotechnology, 2020), GATE XL (All India Rank 8, 2019), DBT-BCIL Fellowship (2019), ICAR AICE-JRF/SRF (All India Rank 3, 2019), and ICAR AIEEA-PG (All India Rank 1, 2017)

Selected Presentations

- **Talk** – "Mechanical control of an epigenetic switch: Mechanoregulation of DNMT3a during wound healing" at *ICMMB 2026: International Conference on Mechanics in Medicine and Biology*, Academia, SGH Campus, Singapore, 2026
- **Poster** – "Mechanoregulation of DNMT3a in the cutaneous wound healing response" at *Mechanobiology in Translational Research: Bioforces to Bedside*, Department of Bioengineering, Indian Institute of Science (IISc), Bengaluru, India, 2025
- **Poster** – "Epigenetic and mechanical regulation of the cutaneous wound healing response" at *Mechanobiology in Health and Disease Conference*, National University of Singapore, Singapore, 2023
- **Poster** – "Role of DNMT3a in the cutaneous wound healing response using a mouse model" at *10th International Conference of Laboratory Animal Scientists' Association (LASA)*, Hyderabad, India, 2022
- **Poster** – "Identification and characterization of stress-responsive PM19 promoter from wheat" at *National Agricultural Science Congress,* New Delhi, India, 2019

Technical Expertise

Experimental Biology

- Experimental design and hypothesis-driven research
- Primary epidermal keratinocyte culture, mouse models, and colony management
- Quantitative live-cell and fluorescence microscopy (confocal and multiphoton), tissue imaging, and flow cytometry (FACS)
- Molecular and cellular biology: genetic perturbation, transfection, qPCR, Western blotting, pseudovirus generation and assays, BSL-2 workflows

Computational & Analytical

- GraphPad Prism, Adobe Illustrator, MATLAB, R (basic), and Python (basic)
- Image analysis (FIJI/ImageJ, CellSense)
- Web development (HTML/CSS/JavaScript) – built and maintain a personal academic website (johanajnabi.com)

Mentoring & Collaboration

- Mentored junior researchers in primary cell culture, molecular biology techniques, and experimental design, including training in data interpretation and scientific communication (presentations and writing)
- Collaborated across interdisciplinary projects spanning mechanobiology, immunology, and spatial transcriptomics, integrating experimental and clinical datasets
- Contributed to multi-author studies across epithelial biology, fibrosis, and host–pathogen interactions, supporting experimental design, data generation, and analysis

Languages

- Bengali (Native)
- English (Advanced)
- Hindi (Limited working proficiency)

References

Prof. Colin Jamora
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Department of Life Sciences, Shiv Nadar Institution of Eminence Deemed to be University, Gautam Buddha Nagar, Tehsil Dadri, Uttar Pradesh – 201314, India
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